

Gabrielle Johnson's Co-Design Canvas — HRI: Check-in Kiosk and Patient Watch

1. Problem

The workflow. Re-triage is already an expected duty, the triage nurse is meant to scan the room at intervals and escalate anyone whose condition has changed. On a 170-patient day, with one nurse, that sweep competes directly with new arrivals.

The specific failure this system targets: deterioration during the wait. A patient triaged ESI 3 who becomes ESI 2 while seated is currently found only if someone happens to notice or the patient returns to the desk.

Two further problems this addresses. Vitals currently consume nurse time inside a 3–5 minute assessment — a worn device captures them before she arrives. And patients repeatedly interrupt staff to ask how much longer, in a room where they can see activity but not progress.

Users.

	Group	Characteristics	Needs
Primary	Patients	Unwell, anxious, in pain; residents and seasonal visitors; wide range of age, language and digital literacy; some confused, intoxicated, blind, deaf, or temporarily unable to read	To be re-checked if they get worse; to know they have not been forgotten; not to be excluded by a device they cannot operate
Secondary	Triage nurses	Mobile; first contact with a frightened/vulnerable patient	Vitals already captured; deterioration surfaced without a manual sweep

Goals.

	Short-term	Long-term
Patients	Be measured without waiting for a nurse; know where they stand	Not come to harm while waiting
Nurses	Deterioration surfaced automatically instead of by eye	The waiting-room sweep becomes a targeted visit rather than a scan

What the system does not do. It does not tell the patient anything clinical, or replace the nurse's own visual sweep. It is a capture and vigilance layer.

Advantages over a person or a fixed screen. Continuous sensing rather than a single reading can predict deterioration better than manual vitals.

2. Ethics

Area	View	Problem	Solution
Physical safety	User	The watch sits on a limb that may have a wound, dressing etc	Contraindication check at issue; staff will apply and remove when necessary.
	System	Unreliable readings from watch.	The watch is passive sensing only; vitals may also be taken by a nurse with a validated device if necessary
Public disclosure	User	A visible alert state tells everyone in the waiting room that this person is deteriorating, and tells the patient before a clinician arrives	Patient facing wording is calm and non-diagnostic eg. "a nurse is coming to check on you"

		— fear itself raises HR and RR, worsening the signal that triggered it	
Sensory accessibility	User	A visual-only signal excludes blind and low vision patients; deaf patients cannot hear their name called; and many ED patients are temporarily impaired, or in too much pain to read	Two or three vibration patterns; every state carries an icon, a word and a motion pattern
	System	An ambient display is useless to a patient who cannot see it	The watch display is never the safety-critical channel therefore, deterioration alerts go to the nurse's handheld device with the patient's location. A patient who cannot see the watch loses convenience, not safety
Access and equity	User	Self check-in excludes those who are older, confused, in pain or not literate in the interface language.	Both routes feed one acuity-ordered queue with neither faster; red-flag presentations bypass the kiosk entirely

	System	Pulse oximetry overestimates SpO ₂ in darker skin	Devices worn by patients must be tested and calibrated before full deployment
Tracking and consent	User	Continuous vitals and indoor location being captured; a QR code on a watch face can be photographed by a bystander	Told plainly at issue what is captured and why, location discarded after the encounter and never stored in a database, short lived rotating QR tokens
	System	The system is in a uniquely rich collection position — passive, continuous, location-aware — and is the first structured patient store Mercer has held, with no existing governance around it	Governance agreed before deployment, data is never stored in a publicly accessible storage space
Physical hygiene	User	A device worn by one unwell patient is then worn by the next	Disinfection between patients, a documented cleaning cadence, hypoallergenic strap material, and a size range so it is not worn too tight
	System	Cleaning time is a staffing cost that competes	Self cleaning or automated cleaning systems

		with the time the system saves	can be implemented alongside system implementation
Unattended failure	User	A worn device fails on a patient, with no clinician to notice, a slipped sensor or a spuriously normal reading creates a false impression of vigilance	Patients told explicitly the watch does not replace telling staff if they feel worse; watches self-test and report their own status to the nurse's queue view
	System	Silent failure is more dangerous than no monitoring	A watch that stops reporting raises its own alert; the nurse's visual sweep is explicitly retained, not replaced

3. Design Guidelines

Advantage guidelines. The design should capitalise on the distinctive strengths of wearable technology: persistence, continuous monitoring and simultaneous physical contact with patients throughout the waiting period. It should not be used to provide reassurance, explanation or clinical judgement, as these functions require human expertise and justify the nurse's continued presence in the waiting area.

Ethical guidelines. Because the device is worn visibly in a public setting, all displayed information must protect the patient's privacy and safety.

Environment guidelines- Patients in this setting may be vulnerable, distressed or temporarily impaired. The patient facing interface should therefore prioritise support and accessibility, whereas the nurse facing interface should prioritise efficiency and accuracy. Patients who cannot use the interface must experience no reduction in monitoring, information or care.

Interaction guidelines. Multimodal- Visual and haptic channels should communicate the same state, ensuring that information remains accessible if either channel is unavailable.

Behaviour guidelines- The watch responds to each patient's baseline rather than relying solely on fixed thresholds, and operates only while the patient remains in the department.

Form guidelines- The kiosk should read as a check-in point, not a diagnostic machine. The watch should look like a wristband, not a medical monitor: an object a frightened person will accept on their body without it becoming a symbol of how sick they might be.

4. MVP

Where and when. Kiosk fixed at the entrance to the waiting area; watch travels with the patient from check-in until they are admitted. 24/7 monitoring, with loads rising after long weekends and through the tourist high season.

Role. A silent chaperone; it accompanies and keeps check on patient vitals

Personality. None, deliberately. No name, no face, no voice, no emotional states, no greeting. Neutral, quiet, and forgettable on the wrist.

Context-based behaviour. Adapts on the patient's own vitals trend rather than population thresholds and wait times relative to their assigned ESI target.

Connection to systems. BLE/UWB anchors for indoor position; the nurse's handheld device and the charge nurse's handheld device for alerts and escalation; the queue engine; the local structured store. No electronic patient record exists at Mercer, so the system stands alone and the paper front page remains the legal record.

Interaction modalities.

	Modalities used
Input	Touch (kiosk taps; wrist contact for sensing), movement (indoor location), screens (kiosk QR scan)
Output	Lights (ambient amber attention state), touch (haptic vibration patterns) screens (kiosk, watch face)

Interaction flow — deterioration during the wait.

	Before	During	After
Patient	Wears the watch; may scan at the kiosk for a position update	Feels a distinct vibration; sees an amber attention state and the words “a nurse is coming to check on you”. Does nothing is	Is seen, reassured, re-triaged or escalated by a person

		never asked to self-assess	
System (watch and kiosk)	Samples HR, SpO ₂ , RR and temperature against the patient's own baseline; reports its own health	Detects a trend change; raises the ambient state locally and sends the clinical detail and location privately to the nurse's handheld	Returns to the neutral waiting state once a nurse acknowledges; logs the episode
Nurse	Carries the handheld device;	Receives the private alert, acknowledges or hands off, pauses her current assessment if needed	Attends the patient, re-triages if required, and the new level re-enters the queue

5. Environment

Where	Mercer ED waiting room. Location coverage must extend to the fast-track area, corridors, toilets and immediately outside, where patients go. Kiosk siting needs disabled access.
When	24/7
Users	Every patient from check-in until they are admitted
Secondary users	Triage nurses and the charge nurse who receive alerts; registration clerks and the greeter who issue watches and route patients who cannot use the kiosk
Simultaneous users	Several, bounded by the room — roughly 70 monitored patients at once, with surges beyond it on heavy days. One kiosk serving both check-in and status checks must never queue
Data collection	vitals, indoor location, chief complaint, arrival mode, age, gender.

External sensors and actuators	<i>Sensors:</i> optical HR and SpO ₂ — degraded by cold hands, poor perfusion, tremor, tattoos, nail varnish and loose fit, all common here and correlated with acuity; skin temperature; BLE or UWB anchors, which are an estates and cabling project rather than a software setting. <i>Actuators:</i> haptic motor and an ambient light state. No motors that move or grip
Connection to systems	Nurse and charge nurse handheld devices; queue; local storage. Must tolerate network loss without losing captured vitals, and fall back to conventional triage when unavailable

6. Form

The watch

Dimension	Position	Rationale
Appearance	Machine-like, abstract — a plain band with a small screen. No face, no character, no anthropomorphism	Patients in the ED are already in distress and vulnerable, hence any device introduced must be familiar and not bothersome
Size	Smaller than human-size — wrist-worn, low profile	Must be forgettable; anything bulky becomes a reminder
Character of movement	Machine-like — no animation; the ambient state is steady, never flashing	Flashing reads as emergency and would broadcast severity to the room
Mobility	Moves with the patient; does not move itself	Deliberately not a mobile robot: a vibration on the wrist

		reaches the patient instantly
Voice and sounds	None. No speech, no beeps, no alarm tones	Audio in a shared room discloses status to strangers and fails for deaf patients; the same information is carried by haptics
Visual cues	One state at a time, shown as icon + word + steady colour. Amber for attention, teal for neutral. Never a number, never an ESI level, never a reason	Patients with different visual impairments should not be impacted/excluded
Touch and smell	Smooth, non-porous, disinfection-tolerant; hypoallergenic strap; neutral to warm, never cold on the skin; no fragrance	Worn against the skin of an unwell person and passed between patients — hygiene and comfort are clinical requirements

The kiosk

Dimension	Position	Rationale
Appearance	Machine-like but approachable — rounded, matte, in the hospital's own palette rather than clinical white	It should read as a check-in point, not a diagnostic device that might deliver bad news
Size	Human-size , with an adjustable screen height	Must serve a wheelchair user and someone who cannot stand for long, without a separate "accessible" version
Character of movement	Static — no moving parts beyond the screen	Nothing that could trap or startle vulnerable patients
Mobility	Fixed, floor-standing, bolted	Predictable location matters more than flexibility; patients

		must be able to find it again
Voice and sounds	Silent by default;	Spoken status in a shared room is a disclosure problem
Touch and smell	Wipeable surfaces, tactile physical buttons alongside touch targets, no fragrance	Tactile keys serve low-vision users and gloved or shaking hands